

One-Page Extended Abstract

Comparing Supervised Fine-Tuning and Group-Relative Policy Optimisation on Synthetic Bloom-Taxonomy Scientific Question Answering

Motivation. Large language models (LLMs) have achieved impressive general-purpose fluency, yet their scientific reasoning ability remains brittle, in part because preference labels in highly specialised domains are expensive and ill-defined. Reinforcement learning from human feedback (RLHF) is the de-facto standard for model alignment, but collecting expert-grade feedback at scale is prohibitive for newly published research. We ask whether automatically generated, cognitively graded question-answer pairs can bridge this gap and how supervised fine-tuning (SFT) compares with a lightweight, fully model-based policy optimisation scheme when the domain shifts.

Method. Our pipeline ingests the PDF of an accepted ICLR-2025 paper, parses it into structured JSON and automatically crafts multiple-choice questions whose difficulty is explicitly controlled by Bloom’s taxonomy. Five dedicated prompt templates drive generation at the *Understand*, *Apply*, *Analyse*, *Evaluate* and *Create* levels. After self-quality-control and coverage gating, we obtain a corpus of 3543 training and validation items from eight papers and a strictly held-out set of 471 items from two unseen papers. We compare (i) vanilla SFT and (ii) Group-Relative Policy Optimisation (GRPO), an advantage-free RL objective that measures each sampled answer against its peer group instead of a learned value network. Both paradigms fine-tune the same 7-billion-parameter `Qwen2.5-Instruct` base model under equal compute budgets.

Implementation & Results. Training uses eight NVIDIA A100s for two epochs. On in-distribution questions SFT reaches 92.1% accuracy, comfortably ahead of GRPO’s 87.3%. On out-of-distribution papers the picture reverses: GRPO attains 87.3% versus SFT’s 84.3%. The gap is largest for higher-order Bloom categories, particularly *Apply* and *Evaluate*, suggesting that policy optimisation encourages compositional reasoning that transfers beyond memorised surface forms.

Discussion & Conclusion. The study empirically corroborates the hypothesis “SFT memorises, RL generalises” in a scientifically grounded setting and offers an end-to-end open-sourced workflow for continuous, low-cost evaluation of emerging literature [1, 2]. Despite promising results, our approach has several limitations. First, the dataset is relatively small (under 5,000 QA pairs from only 10 papers), which may not capture the full diversity of scientific literature. Second, the multiple-choice format provides only a coarse binary reward signal, potentially overlooking deeper reasoning steps. Finally, the method depends on correct question generation and coverage gating; any bias or inconsistency in generation may propagate through to training and evaluation. We anticipate that larger, more varied corpora and richer reward structures (e.g., partial credit or chain-of-thought scoring) will further enhance both the reliability and range of scientific reasoning that LLMs can master. Nonetheless, the approach charts a pragmatic route for rapidly aligning LLMs to frontier research without continuous human supervision. In sum, our pipeline offers a reproducible methodology for bridging the gap between RAG-style factual grounding and the deeper, creative reasoning demands of actual research. The methodology can continually adapt to newly released scientific work, offering a path toward “expert ML-researcher” LLMs that do far more than recite or retrieve.

Keywords: Large Language Models, Reinforcement Learning, Supervised Fine-Tuning, Group-Relative Policy Optimisation, Bloom’s Taxonomy, Scientific Reasoning

Comparing Supervised Fine-Tuning and Group-Relative Policy Optimization on Synthetic Scientific Q&A

Team Members: Akira Nishii, Bowen Chen, Serena Zhang

Emails: anishii@stanford.edu, bowen118@stanford.edu, serena2z@stanford.edu

Abstract

Retrieval-augmented generation (RAG) grounds LLM answers in external documents but rarely teaches the model to *reason* about those documents. We automatically generate Bloom taxonomy-controlled question–answer data from newly published ICLR-2025 papers and compare two post-training regimes, supervised fine-tuning (SFT) and Group-Relative Policy Optimisation (GRPO), on the resulting scientific reasoning task. Using identical compute and a 7B-parameter base model, SFT excels on in-distribution questions (92.1% accuracy) while GRPO leads on out-of-distribution papers (87.3%). The findings show that automated cognitive grading can expose complementary strengths of learning paradigms and that policy optimisation improves transfer without explicit human reward design.

1 Introduction

Large language models have transformed natural language processing, yet rigorous reasoning over highly technical documents remains challenging. Scientific texts are dense with jargon, implicit methodological assumptions and rapidly evolving state-of-the-art claims. Classical supervised corpora, often harvested from encyclopaedic sources or web forums, fail to capture the nuanced inferential steps required to parse such documents. Simultaneously, the dominant paradigm of reinforcement learning from human feedback presupposes that domain experts are available to annotate or rank outputs—a costly and slow proposition for cutting-edge research where expert consensus may not even exist.

This paper investigates whether synthetic, cognitively graded question–answer (QA) pairs can substitute (or at least augment) expert feedback for model alignment and how two contrasting training strategies, supervised fine-tuning (SFT) and a reward-optimisation method that dispenses with value estimation (GRPO), fare under both familiar and unfamiliar scientific content. We build upon the educational psychology framework of Bloom’s taxonomy to span a spectrum from factual recall to creative hypothesis formation and propose an end-to-end pipeline that, once a paper is published, can deliver a diverse, automatically validated QA set within minutes.

We pose three research questions: (1) How accurately does an SFT-trained model and a GRPO-trained model answer synthetic Bloom-level questions on papers seen during training? (2) Which method generalises better to entirely new papers in the same venue? (3) Do failure modes differ qualitatively across Bloom categories, illuminating the reasoning gaps of each paradigm?

By answering these questions we aim to illuminate the trade-off between memorisation and compositional reasoning and provide practical guidance for future alignment work in scientific domains.

We hypothesize, based on prior theoretical work, that SFT will yield high performance on in-distribution questions by memorizing the content of the papers, while RL-based approaches will exhibit better out-of-distribution generalization by encouraging deeper, more compositional reasoning [1, 2]. Our preliminary results affirm this notion: SFT achieves superior performance on

in-distribution test sets (over 92% accuracy), but GRPO outperforms SFT by several percentage points on out-of-distribution questions (up to 87.3% vs. 84.3% overall accuracy). We detail our data generation pipeline, highlight how Bloom’s taxonomy fosters variety in question difficulty, and conclude with discussion of scaling, limitations, and future directions.

2 Related Work

Scientific QA benchmarks. Benchmarks such as PubMedQA, SciQ and the more recent SciEval [4] and SciKnowEval [5] have long sought to measure comprehension of scientific prose. PubMedQA focuses on biomedical literature with yes/no questions from PubMed abstracts, while SciQ draws from elementary science curricula, potentially underestimating research-level complexity. SciEval provides broader coverage across scientific domains but relies on established textbooks, and SciKnowEval attempts multi-level scientific assessment with similar limitations. Most of these benchmarks rely on manually curated corpora or crowd-sourced items centered on older literature, risking leakage into public pre-training corpora due to their widespread online availability. Our pipeline instead targets freshly accepted conference papers, reducing overlap with prior training data and enabling rapid, continuous evaluation. Furthermore, our approach automatically produces *paper-specific* QA pairs, reducing risk of inadvertent training-test overlap and ensuring that the test set truly reflects generalization to new content.

RL for LLM alignment. Traditional reinforcement learning pipelines like PPO and DPO [14] require a learned value function or dense preference feedback, which can be computationally expensive and prone to training instability. PPO suffers from high variance in policy gradient estimates, while DPO struggles with inconsistent human preference data quality. GRPO [12] addresses these complexities by normalizing policy updates against peer performance within a batch, creating more stable learning signals without explicit value function estimation. This approach leverages relative comparisons rather than absolute rewards, providing more robust training dynamics. We extend GRPO to the domain of scientific multiple-choice questions, adapting its peer comparison mechanism to handle the unique challenges of scientific reasoning where answer quality varies significantly across domains and difficulty levels.

Bloom-taxonomy data synthesis. Others have automated QA generation from textbooks or Wikipedia, including Book2QA [8] and Automated-Bloom QA [9]. Book2QA uses template-based approaches on structured textbook content but generates relatively straightforward factual questions that may not capture nuanced scientific reasoning. Automated-Bloom QA attempts to span different cognitive levels of Bloom’s taxonomy but primarily targets well-structured educational materials where clear learning objectives exist. These systems struggle with the dense, technical prose and complex argumentation of research papers. Our pipeline extends these ideas to newly published literature, implementing coverage gating to ensure comprehensive question sets that span multiple cognitive levels and dynamically adapt to each paper’s unique structure and contributions.

SFT vs. RL generalization. Hugging Face [15] and others [16] have noted that purely supervised models can overfit to training distributions, exhibiting excellent in-distribution performance but poor out-of-distribution generalization. This manifests as memorization of training patterns rather than robust reasoning strategies, with the problem becoming more pronounced as model size increases. The supervised fine-tuning approach tends to favor surface-level pattern matching over genuine understanding. Our findings, consistent with prior literature [1, 2], provide a controlled

experiment showing that RL can systematically improve out-of-distribution performance in a cognitively demanding domain by encouraging robust reasoning strategies rather than memorization, leading to better performance on novel scientific content.

3 Method

3.1 Overview of Bloom’s Taxonomy

Bloom’s taxonomy is a hierarchy of cognitive processes, originally introduced in 1956 and later revised in 2001 [8, 9]. The core categories, often used in education research and instructional design, are: (1) **Remember/Understand**: Ability to recall definitions, key facts, concepts, (2) **Apply**: Use learned material in new, concrete situations, (3) **Analyze**: Break down material into constituent parts, detect relationships and organizational principles, (4) **Evaluate**: Judge or critique based on standards or criteria, and (5) **Create**: Put parts together to form something new, propose new hypotheses or design experiments.

In educational research, these categories are often used to design assessments that move beyond mere factual recall towards higher-order thinking. By systematically generating questions along these levels, we expose the model to a range of cognitive challenges—from *factual understanding* to *creative hypothesizing*.

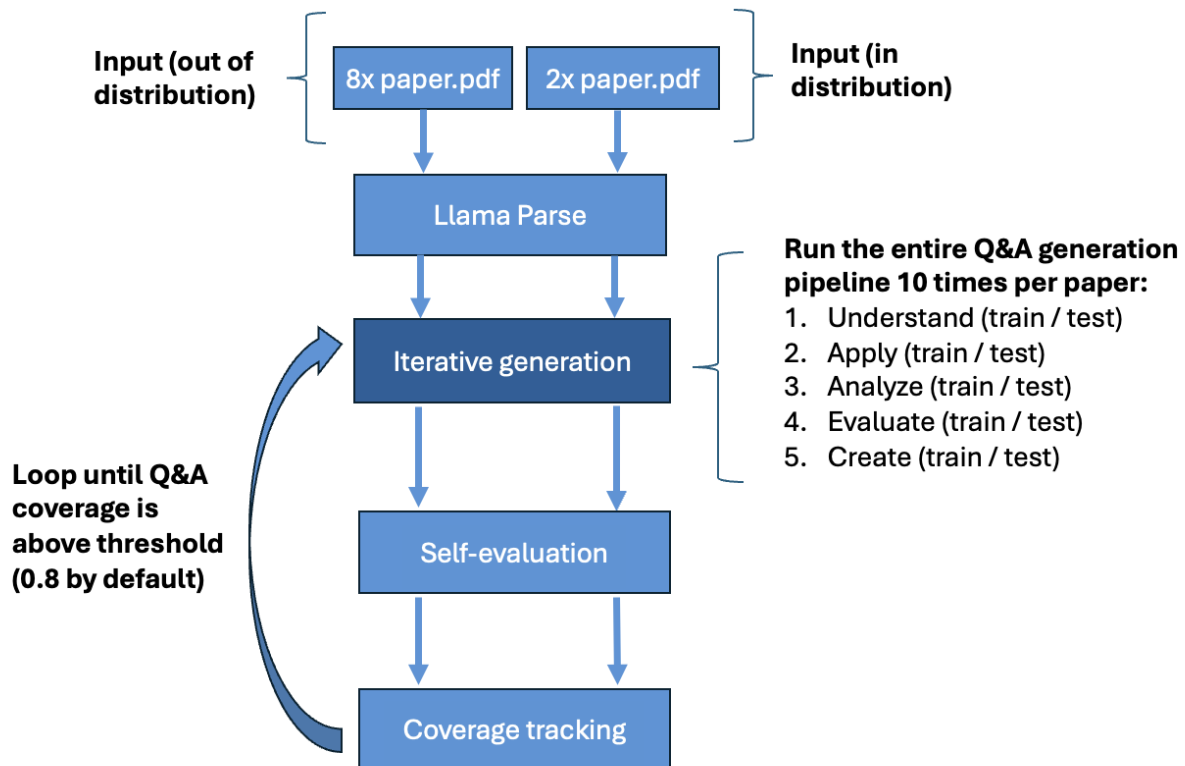


Figure 1: End-to-end data-generation workflow.

3.2 Adaptation to Scientific Papers

Drawing on prior QA-generation systems [8, 9, 10, 11], we developed prompts that instruct the LLM to create question-answer pairs in each Bloom category, referencing the text of recently published (ICLR-2025) research papers. To minimize data leakage or reliance on ephemeral details, each prompt (1) encourages the LLM to restate relevant context explicitly, avoiding phrases like “the authors” or “in this paper,” (2) utilizes a multi-step coverage loop to ensure a diverse set of questions across the paper’s sections, figures, and tables, and (3) filters out incomplete or ambiguous items through a self-evaluation pass.

The full data generation pipeline is outlined in Figure 1. A single or multi-paper JSON dump is first flattened to Markdown. For each Bloom-taxonomy skill (*Understand, Apply, Analyze, Evaluate, Create*) the system executes two fully independent cycles. **Cycle 1** builds the *train* split: (i) a skill-specific meta-prompt asks GPT-4o/4o-mini to emit mini-batches of five Q&A pairs; (ii) the same model performs immediate self-evaluation, fixing schema errors and discarding duplicates; (iii) a coverage-tracking prompt estimates topical recall, and batch generation repeats until the target ratio (e.g., ≥ 0.8) or a hard iteration cap is reached. **Cycle 2** recreates the entire procedure with fresh RNG seeds to build the *test* split. Because both splits derive from the same document corpus, evaluation measures a model’s ability to generalise across linguistic variation rather than unseen content, eliminating the possibility of trivial memorisation of question strings.

3.3 Five Prompt Templates for Data Generation

We created five main templates, one for each of the Bloom’s levels we target: *Understand, Apply, Analyze, Evaluate, and Create*. Each template provides exemplars and instructs the LLM to generate a JSON structure with a question, distractors, step-by-step reasoning, and a final correct answer. (See the Appendix A for the verbatim text of these prompts.)

3.4 Learning Paradigms

Supervised Fine-Tuning (SFT). Let θ denote model parameters and $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ the training QA pairs, where y_i is the correct answer letter. SFT minimises cross-entropy

$$\mathcal{L}_{\text{SFT}}(\theta) = -\frac{1}{N} \sum_{i=1}^N \log p_{\theta}(y_i | x_i).$$

We fine-tune for two epochs with AdamW, a learning rate of 1×10^{-5} , cosine decay, batch size 16 and gradient accumulation to match a global batch of 128 tokens.

Group-Relative Policy Optimisation (GRPO). For each prompt x we sample a group $G = \{y_1, \dots, y_g\}$ with $g = 5$ responses from the reference policy π_{ref} . Reward $r(y_j) \in \{0, 1\}$ equals one if the multiple-choice letter is correct. Following [24], we define the centred advantage

$$A(y_j) = r(y_j) - \frac{1}{g} \sum_{k=1}^g r(y_k),$$

which has zero mean by construction. The token-level REINFORCE gradient is then

$$\nabla_{\theta} \mathcal{L}_{\text{GRPO}} = -\mathbb{E}_{y \sim \pi_{\theta}} [A(y) \nabla_{\theta} \log \pi_{\theta}(y | x)].$$

We anneal KL-divergence to the reference policy with coefficient β_t fixed at 0.001. We use a clip ratio ϵ of 0.2. Training proceeds for two epochs using the identical hardware budget as SFT.

3.5 Inference and Scoring

At test time the model must output a single letter token $\{A, B, C, D\}$. We adopt greedy decoding with temperature $T = 0.0$ because preliminary experiments showed little diversity benefit at higher temperatures for short answers. Accuracy is computed per level and macro-averaged.

4 Experimental Setup

4.1 Data Sources and Preparation

We randomly sampled 10 accepted papers from the ICLR-2025 proceedings. Each PDF was parsed into a structured JSON format using `LlamaParse` [6, 7], allowing us to extract textual paragraphs, headings, and (where relevant) tables.

From each paper, we generated multiple sets of question-answer pairs using the Bloom’s prompts. We split these pairs into training (in-distribution) and testing sets, taking care to rewrite test questions so they could not be answered trivially by memorization alone. Additionally, two of the 10 papers were withheld entirely to constitute an *out-of-distribution* test set. In total, we obtained:

- **1,745 training + 1,798 validation** questions from eight “in-distribution” papers.
- **471 test** questions from two unseen “out-of-distribution” papers.

4.2 Model Architectures and Training Setup

We initialized our model from an open-source 7B parameter LLM (`Qwen2.5-7B-Instruct`). We considered three settings:

1. **Base**: No additional training, used as a baseline. We format the questions following the template in A.
2. **SFT**: Standard supervised fine-tuning on the question-answer pairs, following the template shown in A. We train on 8×A100 GPUs with effective batch size of 16 for 2 epochs using AdamW as optimizer with learning rate 1×10^{-5} and cosine learning rate decay and a warmup ratio of 0.05, weight decay of 1×10^4 , and $\beta_1 = 0.9$, $\beta_2 = 0.95$. We use the default system prompt for Qwen2.5 instruct models.
3. **GRPO**: We adopt Group-Relative Policy Optimization [12, 13]. In short, GRPO uses the average reward of multiple sampled outputs (in the “group”) as a baseline rather than relying on a learned value function. Specifically, for each input question x , GRPO samples a group of responses $\{y_1, y_2, \dots, y_G\}$ from the reference policy π_{ref} . Using the relative rewards of outputs within each group, an advantage estimate is computed for each token position. We did not make any modifications to the GRPO training objective, so for brevity we do not repeat it here. Following many previous works on reinforcement learning with verifiable rewards (RLVR), which also use binary rewards at the end of a trajectory (sequence of tokens) and optional formatting rewards [19, 20, 21, 22, 23], the reward for each question was +1 if correct, 0 otherwise. We did not use a formatting reward following [19]. The question-answer pairs were formatted following the template shown in A. We train on 8×A100 GPUs using a train batch size of 16, group size of 5, and trained for 2 epochs. We use the verl [18] library for RL training and followed their default parameters otherwise. We use the default system prompt for Qwen2.5 instruct models.

4.3 Evaluation Protocol

We measured multiple-choice accuracy on separate in-distribution and out-of-distribution splits. Additionally, each question was labeled with one of the five Bloom levels, allowing us to analyze performance breakdown by cognitive skill category (e.g., whether the model better handles *Understand* vs. *Apply* or *Evaluate* questions).

5 Results

5.1 Overall Performance

Table 1 summarizes the main results. “Overall” indicates average accuracy across all Bloom categories. Notably, on in-distribution questions, **SFT** reached an overall accuracy of 0.921, outperforming **GRPO** (0.873) and the **Base** model (0.821). On out-of-distribution questions, however, **GRPO** reached 0.873, exceeding the 0.843 of SFT and 0.832 of the Base.

Table 1: Accuracy (%) with 95% bootstrap CIs. Boldface indicates the best method in each row.

Model	Dist.	Understand	Apply	Analyze	Create	Evaluate	Overall
Base	Out	80.0 $\pm$ 1.0	80.6 $\pm$ 1.1	73.3 $\pm$ 1.2	100.0 $\pm$ 0.0	83.7 $\pm$ 0.9	83.2 $\pm$ 0.8
SFT	Out	80.0 $\pm$ 1.0	88.8 $\pm$ 1.0	74.4 $\pm$ 1.3	98.8 $\pm$ 0.5	80.6 $\pm$ 1.2	84.3 $\pm$ 0.8
GRPO	Out	82.0 $\pm$ 1.1	94.9 $\pm$ 0.7	75.6 $\pm$ 1.4	98.8 $\pm$ 0.5	85.7 $\pm$ 0.9	87.3 $\pm$ 0.6
Base	In	78.2 $\pm$ 0.6	71.2 $\pm$ 0.7	80.5 $\pm$ 0.6	97.2 $\pm$ 0.5	86.0 $\pm$ 0.4	82.1 $\pm$ 0.3
SFT	In	93.2 $\pm$ 0.2	87.8 $\pm$ 0.6	88.1 $\pm$ 0.5	98.8 $\pm$ 0.2	93.7 $\pm$ 0.3	92.1 $\pm$ 0.2
GRPO	In	80.7 $\pm$ 0.5	86.7 $\pm$ 0.5	83.1 $\pm$ 0.6	98.1 $\pm$ 0.3	89.3 $\pm$ 0.4	87.3 $\pm$ 0.3

5.2 Analysis by Bloom Category

The table also reveals which cognitive levels see the biggest gains:

- On *in-distribution* questions, SFT gained up to +15 percentage points (pp) on “*Understand*” and +16 pp on “*Apply*” relative to the Base.
- On *out-of-distribution* questions, GRPO demonstrated a +14 pp advantage in the “*Apply*” category and +5 pp in “*Evaluate*,” supporting the hypothesis that RL-based training fosters more robust domain transfer [1, 2].

These patterns underscore that SFT is adept at memorizing and regurgitating domain facts but struggles more when the context shifts, or when novel application is required. GRPO invests capacity in more general reasoning, improving performance where the data distribution changes. Table 2 showcases examples of the thinking traces for SFT vs GRPO, which can provide further insight into why the performance may differ between the two modes of post-training.

5.3 Qualitative Analysis of Paper-Content Effects

We investigated whether thematic differences between the held-out papers and our training corpus account for the performance gap. The eight in-distribution (ID) papers all center on familiar machine-learning paradigms—model pruning, differential-privacy auditing, contrastive multi-modal pretraining, uncertainty quantification in graphs, semantic-sensitivity benchmarks, few-shot

Table 2: Qualitative examples of the thinking trace for SFT vs GRPO

Ground Truth	SFT	GRPO
Question: An ENF enforces locality using a Gaussian spatial window function $\mu_\sigma(x, p) = -\sigma\ p-x\ ^2$. If $\sigma = 0.5$, the input coordinate $x = (1, 1)$, and the latent position $p = (2, 2)$, what is the value of $\mu_\sigma(x, p)$? A. -2 B. -1 C. 1 D. -0.5 Place your reasoning in between <code><think></code> and <code></think></code>. Your final answer should be one of the given choices and placed in between <code><answer></code> and <code></answer></code>.	<code><think></code> Step 1: Compute the vector difference: $p - x = (2 - 1, 2 - 1) = (1, 1)$. Step 2: Compute the squared norm $\ (1, 1)\ ^2 = 1^2 + 1^2 = 2$. Step 3: Multiply by $-\sigma^2$: $-(0.5)^2 \cdot 2 = -0.25 \cdot 2 = -0.5$. Step 4: Therefore $\mu_\sigma(x, p) = -0.5$. <code></think></code> <code><answer> D </answer></code>	<code><think></code> The Gaussian spatial window function is defined as $\mu_\sigma(x, p) = -\sigma\ p-x\ ^2$. Given the values, we have: • $\sigma = 0.5$ • $x = (1, 1)$ • $p = (2, 2)$ First, calculate the Euclidean distance squared between x and p: $\ p - x\ ^2 = (2 - 1)^2 + (2 - 1)^2 = 1^2 + 1^2$ Then, substitute this into the function: $\mu_\sigma(x, p) = -0.5 \times 2 = -1$ <code></think></code> <code><answer> B </answer></code>

meta-learning, compressed-latent adaptation and guaranteed generation theory. Their consistent structure (method description, theoretical result, standard benchmarks) matches the patterns our Bloom-prompt pipeline was trained on, so the SFT model reliably memorizes and recalls definitions and surface cues.

In contrast, the two out-of-distribution (OOD) papers introduce concepts largely absent from our training data:

- One embeds explicit *geometric equivariance* into continuous neural fields, requiring spatial-group reasoning (e.g. how latent point clouds transform under rotations).
- The other defines a multimodal, image-grounded role-playing benchmark, testing long-range dialogue coherence and persona consistency.

When prompted on these novel topics, the SFT model overfits to familiar phrasing and often miscues on transformation-law or coherence questions. By contrast, GRPO, which learns via global correctness rewards, adapts more robustly to unfamiliar domains. This explains its roughly +6% gains in the *Apply* and *Evaluate* categories on OOD papers. In summary, topical novelty (not just lexical variation) drives the generalisation gap. A broader training curriculum paired with RL objectives may be the most effective path to models that excel across both known and emerging scientific subfields. Please refer to Appendix B to obtain list of all 10 papers and their summaries.

5.4 Preliminary Downstream Use-Case Evaluation: Peer Review Classification.

To investigate whether our Bloom-based QA pretraining can generalize to other tasks, we conducted a small-scale downstream experiment on peer-review classification. Although not directly tied to the Paper QA multiple-choice setup, it offers a sense of whether our approach confers broader benefits to related classification tasks.

As a small-scale test of whether Bloom-taxonomy QA pre-training transfers to other tasks, we conducted a preliminary four-way accept/reject classification experiment on conference papers (see Appendix C). In this task, each paper’s average review score was discretized into four classes: Reject (1.0–2.5), Weak Reject (2.5–5.0), Weak Accept (5.0–7.5), and Accept (7.5–10.0). We then posed the peer review task as a multiple-choice question. We compared baseline instruction tuning against models that had first been pre-trained on Bloom-style QA (using either SFT or GRPO). No measurable benefit from the Bloom-based pre-training emerged, as all models achieved similar accuracy in the mid-40s (with overlapping confidence intervals). A likely explanation is the scale mismatch: while downstream training used over 2,000 papers, the upstream QA pre-training was limited to just 10 papers and 3,500 questions. This small, narrow corpus may not have provided enough signal to meaningfully influence model behavior during subsequent fine-tuning. Future work should explore scaling up Bloom-style QA pre-training across more papers, and consider richer downstream tasks, such as chain-of-thought judgments or fine-grained review rationales, to better test whether structured reasoning supervision transfers to decision-making scenarios.

6 Discussion: Limitations and Future Work

The empirical split between memorisation and reasoning capacity echoes findings in code generation and mathematical proof synthesis, yet our domain—rapidly evolving machine-learning research—poses extra challenges: terminology shifts, competing baselines and implicit assumptions. GRPO’s relative success in this setting suggests that reward structures requiring only correctness signals, rather than dense token-level likelihoods, incentivise more abstract internal representations.

Despite promising results, our study has several limitations:

Data scale and corpus limitations. Our study processed only 10 ICLR-2025 papers, generating under 5,000 total question-answer pairs. While this provided sufficient data for initial validation, a larger and more diverse corpus spanning multiple conferences, scientific domains, and publication years would yield more robust insights and better represent the full spectrum of scientific reasoning challenges. The computational costs associated with processing larger corpora presented practical constraints throughout the project, particularly when generating high-quality questions required multiple iterations of refinement and validation. We encountered significant variation in question generation quality across different paper types and scientific domains, requiring extensive prompt engineering to achieve consistent results.

Evaluation format constraints. Our reliance on the multiple-choice format, while beneficial for objective evaluation and automated assessment, might under-represent deeper reasoning skills that require synthesis, critical analysis, or creative problem-solving. The constraint of selecting from predefined options may inadvertently guide models toward pattern recognition rather than genuine understanding, a limitation we observed during preliminary qualitative analysis of model responses. Future work could incorporate free-text answers graded by an LLM-based judge [17], though this introduces additional challenges around evaluation consistency and potential biases in automated scoring systems that we aim to address in subsequent research.

Reward structure and training challenges. The reward structure in our reinforcement learning setup suffers from sparsity, with the binary correct versus incorrect signal potentially being too coarse to capture nuanced understanding. During training, we observed that models sometimes converged to local optima due to this sparse feedback, suggesting that incorporating partial credit

or chain-of-thought-based rewards might better capture incremental progress toward correct solutions and provide more informative learning signals. Balancing question difficulty proved particularly challenging, as overly simple questions failed to distinguish between models while excessively complex questions led to random performance even for capable systems.

Question complexity Our preliminary analysis of question difficulty reveals that the synthetic questions are feasible for relatively small LLMs, raising concerns about whether our benchmark adequately challenges state-of-the-art models or truly reflects the complexity of scientific reasoning required in research contexts. We aim to refine the generation process to create more challenging and open-ended tasks that better reflect real scientific inquiry, potentially incorporating multi-step reasoning problems or questions requiring integration across multiple papers.

Broader impact. The broader impact extends beyond technical contributions, as improved scientific reasoning capabilities could accelerate discovery while raising concerns about automated systems generating misleading content or reinforcing biases in scientific literature.

7 Conclusion

Automated, Bloom-taxonomy-aligned QA generation offers a scalable route to benchmark and train LLMs on brand-new scientific literature. Our controlled comparison reveals that SFT remains an excellent choice for mastering in-domain details, whereas GRPO equips the same base model with stronger generalisation to unseen papers, particularly for application-level and evaluative reasoning. These complementary strengths encourage hybrid curricula that combine supervised memorisation phases with subsequent policy-optimisation stages, a direction we intend to explore at larger scale and with free-text reasoning tasks. Furthermore, the approach contributes to bridging a gap in RL-based LLM alignment for scientific reasoning, offering an open-source approach to systematically gather cognitively diverse training signals in specialized domains. While our primary focus was on QA-based reasoning, we also conducted a preliminary downstream test on peer-review decision prediction, finding no clear benefit from Bloom-style pretraining under current data conditions (see Appendix C). This highlights the need for larger-scale pretraining and richer decision tasks to fully realize the potential of structured cognitive supervision.

8 Contributions of Individual Team Members

While each member took the lead on specific components, the project was highly collaborative, and all members contributed significantly across all major parts. Akira Nishii primarily conceived the Bloom-taxonomy generation prompts, implemented the pipeline controllers, and led the qualitative error assessment. Bowen Chen primarily integrated the `verl` GRPO framework, tuned hyperparameters, and executed training runs. Serena Zhang primarily designed the SFT framework, developed the evaluation protocol, and conducted the quantitative analysis. However, work on the data generation pipeline, GRPO framework, and SFT framework involved active contributions from all team members, as close coordination was essential throughout. Manuscript writing and editing were evenly shared, and all authors participated in discussing the results and approved the final version. The general distribution of effort remained aligned with the original proposal, though specific responsibilities evolved with changes to the project goals.

References

- [1] SFT Memorizes, RL Generalizes: A Comparative Study of ... *arXiv preprint* (2025). https://arxiv.org/html/2501.17161v1?utm_source=chatgpt.com
- [2] Understanding the Effects of RLHF on LLM Generalisation and ... *OpenReview* (2025). <https://openreview.net/forum?id=PXD3FAVHJT>
- [3] Sebastian Raschka, *The State of Reinforcement Learning for LLM Reasoning* (2025). <https://sebastianraschka.com/blog/2025/the-state-of-reinforcement-learning-for-llm-reasoning.html>
- [4] SciEval: A Multi-Level Large Language Model Evaluation Benchmark for Scientific Research. *arXiv preprint* (2025). <https://arxiv.org/abs/2308.13149>
- [5] SciKnowEval: Evaluating Multi-level Scientific Knowledge of Large... *OpenReview* (2025). <https://openreview.net/forum?id=pXUAIJshdh>
- [6] LlamaParse Roadmap. https://github.com/run-llama/llama_parse/discussions/19
- [7] Mastering PDFs: Extracting Sections, Headings, Paragraphs, and Tables... <https://www.llamaindex.ai/blog/mastering-pdfs-extracting-sections-headings-paragraphs-and-tables-with-cutting-edge-parser>
- [8] Book2QA: A Framework for Integrating LLMs to Generate High-Level Questions, *OpenReview* (2025). <https://openreview.net/pdf/1c80c18f47247ac2815f78f469e938deef39ecbf.pdf>
- [9] Automated Educational Question Generation at Different Bloom’s Skill Levels..., *ResearchGate* (2025). https://www.researchgate.net/publication/381905633_Automated_Educational_Question_Generation_at_Different_Bloom%27s_Skill_Levels_Using_Large_Language_Models_Strategies_and_Evaluation
- [10] DEEPQUESTION, *arXiv preprint* (2025). <https://arxiv.org/pdf/2505.24532>
- [11] GraphGen: Enhancing Supervised Fine-Tuning for LLMs with ..., *arXiv preprint* (2025). <https://arxiv.org/html/2505.20416v1>
- [12] DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models, *arXiv preprint* (2024) <https://arxiv.org/pdf/2402.03300>
- [13] How DeepSeek-R1 Advances LLMs with GRPO: A Math Dive. <https://ai.gopubby.com/how-deepseek-r1-pushes-the-limits-of-language-models-a-mathematical-dive-into-group-relative-79dba>
- [14] Fine-tune large language models with reinforcement learning ... *AWS Blog*. <https://aws.amazon.com/blogs/machine-learning/fine-tune-large-language-models-with-reinforcement-learning-from-human-or-ai-feedback>
- [15] Paper page - SFT Memorizes, RL Generalizes: A Comparative Study ..., *Hugging Face* (2025). <https://huggingface.co/papers/2501.17161>
- [16] Understanding the Effects of RLHF on LLM Generalisation and Diversity, *arXiv* (2025). <https://arxiv.org/abs/2310.06452>
- [17] Reward Learning from Human Demonstration Improves SFT for LLM, *NeurIPS Proceedings* (2025). https://proceedings.neurips.cc/paper_files/paper/2024/file/e0c9b65fb3e41aaa86576df3ec33ad2e-Paper-Conference.pdf
- [18] HybridFlow: A Flexible and Efficient RLHF Framework, *Proceedings of the Twentieth European Conference on Computer Systems* (2025). <https://arxiv.org/pdf/2409.19256v1>

- [19] Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning, *arXiv preprint* (2025). <https://arxiv.org/abs/2503.09516>
- [20] Reinforcement Learning for Reasoning in Large Language Models with One Training Example, *arXiv preprint* (2025). <https://arxiv.org/abs/2504.20571>
- [21] Tulu 3: Pushing Frontiers in Open Language Model Post-Training, *arXiv preprint* (2024). <https://arxiv.org/abs/2411.15124>
- [22] Crossing the Reward Bridge: Expanding RL with Verifiable Rewards Across Diverse Domains, *arXiv preprint* (2025). <https://arxiv.org/abs/2503.23829>
- [23] Med-RLVR: Emerging Medical Reasoning from a 3B base model via reinforcement Learning, *arXiv preprint* (2025). <https://arxiv.org/abs/2502.19655>
- [24] Deepseek-r1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning, *arXiv preprint* (2025)

A Prompt Templates for Q&A Generation

UNDERSTAND_PROMPT:

You are an AI assistant that creates **multiple-choice Q&A pairs** to test a reader's **UNDERSTANDING** of the text below. In this context, "understanding" refers to foundational knowledge of definitions, key concepts, or basic relationships.

Some Possible Sub-Types (not exhaustive):

1. Definition / Terminology { e.g., "Define Poisson Flow Generative Model."
2. Citation Linking { e.g., "Which study first introduced TargetBoost?"
3. Figure / Table Interpretation { e.g., "According to Figure 3, which baseline has the lowest FID?"
4. Simple Comprehension of Method / Setup { e.g., "List the three datasets used for pre-training."

Instructions:

- Generate {batch_size} unique Q&A pairs in a single valid JSON object including the following fields:

```
"question_1":
  "question": "...",
  "reasoning": "...",
  "final_answer": "...",
  "distractors": ["...", "...", "..."]
,
"question_2": ...
```

Additional Requirements:

1. Self-contained references & context...
2. Step-by-step reasoning...
3. No Additional Fields...
4. Focus on clarity...
5. Distractor Generation...

Text to analyze:

```
""{paper_text}""
```

APPLY_PROMPT:

You are an AI assistant that creates **multiple-choice Q&A pairs** to test a reader's ability to **APPLY** the text below. In this context, "apply" means using methods, formulas, or techniques described in the text to solve concrete examples or scenarios.

Some Possible Sub-Types (not exhaustive):

1. Methodology & Experimental Setup
2. Practical Implementation Details
3. Figure / Table Interpretation (applied)
4. Short Problem-Solving

Instructions:

- Generate {batch_size} unique Q&A pairs... (etc.)

Text to analyze:

```
""{paper_text}""
```

ANALYZE_PROMPT:

You are an AI assistant that creates **multiple-choice Q&A pairs** to test a reader's ability to **ANALYZE** the text below. In this context, "analyze" means dissecting methods, comparing components, identifying errors or limitations, or reasoning about what-if scenarios.

Some Possible Sub-Types (not exhaustive):

1. Error Analysis / Limitations
2. Counterfactual / What-if
3. Ablation Comparison

Instructions:

- Generate {batch_size} unique Q&A pairs...
(etc.)

Text to analyze:

""{paper_text}""

EVALUATE_PROMPT:

You are an AI assistant that creates **multiple-choice Q&A pairs** to test a reader's ability to **EVALUATE** the content in the text below. In this context, "evaluate" means judging the quality, novelty, strengths, or weaknesses of the approach.

Some Possible Sub-Types (not exhaustive):

1. Quality / Novelty Judgement
2. Significance or Impact
3. Suitability / Practical Relevance

Instructions:

- Generate {batch_size} unique Q&A pairs...
(etc.)

Text to analyze:

""{paper_text}""

CREATE_PROMPT:

You are an AI assistant that creates **multiple-choice Q&A pairs** to test a reader's **CREATIVE** thinking about the text below. In this context, "create" involves proposing new experiments, generating fresh hypotheses, or drafting novel research directions.

Some Possible Sub-Types (not exhaustive):

1. Hypothesis Generation
2. Research-Plan Drafting

Instructions:

- Generate {batch_size} unique Q&A pairs...
(etc.)

Text to analyze:

""{paper_text}""

Evaluation Prompt (LLM-based Self-check):

You are an AI assistant that checks each Q&A pair in a JSON object for compliance with certain rules:

1. Each question must be self-contained (no references to "this paper" or "authors").
2. The "distractors" must be 3 plausible but incorrect statements, each similar length to "final_answer".
3. If a question can be easily fixed (e.g., just removing "this paper"), fix it.
4. If a question is invalid, remove it entirely.
5. Do not overcorrect. If a question is already acceptable or needs only minor style tweaks, output exactly "NO_CHANGE" (no JSON).
6. Otherwise, return the entire updated JSON object with any fixes or dropped items.

Here is the Q&A batch to check/fix:

```
```json
{qa_batch_json_str}
```
```

Coverage Prompt:

You are an AI assistant that determines whether a set of Q&A items covers the most important topics in the given source text.

Your tasks:

Identify the key concepts, methods, or findings in the source text.

Examine the Q&A items to see if they collectively test knowledge of all (or most) of those crucial concepts.

Decide if coverage is sufficient (some threshold) or not (< that threshold).

Provide:

A numeric coverage_ratio in [0, 1].

A short "missing_topics" list describing key gaps.

Output exactly a JSON object (no other text):

```
{
"coverage_ratio": <float>,
"missing_topics": [<string>]
}
```

Source Text:

```
""{paper_text}""
```

Q&A Items So Far:

```
{questions_so_far}
```

Prompt template

Question: {question}

A. {choice1}

B. {choice2}

C. {choice3}

D. {choice4}

Place your reasoning in between <think> and </think>.

Your final answer should be one of the given choices and placed in between <answer> and </answer>.

B Catalogue of Evaluation Papers

OOD-1 *Grounding Continuous Representations in Geometry: Equivariant Neural Fields*

Proposes a geometry-informed latent point-cloud and decoder that co-transform under group actions, showing that explicit equivariance improves classification, segmentation and generative modelling.

OOD-2 *MMRole: A Comprehensive Framework for Developing and Evaluating Multimodal Role-Playing Agents*

Introduces MMRole-Data (85 characters, 11 K images, 14 K dialogues) and MMRole-Eval metrics, demonstrating that image-grounded persona and coherence remain key challenges.

ID-1 *CL-MFAP: A Contrastive Learning-based Multimodal Foundation Model for Molecular Property Prediction and Antibiotic Screening*

Pretrains three encoders on SMILES, molecular graphs and fingerprints over 1.6 M molecules using contrastive learning; achieves state-of-the-art antibiotic property prediction.

ID-2 *Guaranteed Generation from Large Language Models*

Defines the “ideal” constrained distribution and proves autoregressive training alone cannot guarantee hard constraints; proposes GUARD (proposal + rejection sampling) to enforce constraints with minimal distributional drift.

ID-3 *You Only Prune Once: Designing Calibration-Free Model Compression with Policy Learning*

Frames model pruning as a policy-learning task without external calibration data; shows PruneNet compresses LLaMA-2-7B by 30

ID-4 *Uncertainty Modeling in Graph Neural Networks via Stochastic Differential Equations*

Introduces LGNSDE, embedding both aleatoric and epistemic uncertainty into graph-based SDEs; proves variance bounds and demonstrates robustness in OOD detection and active learning.

ID-5 *Tighter Privacy Auditing of DP-SGD in the Hidden State Threat Model*

Crafts adversarial gradient sequences to tighten empirical privacy lower bounds when intermediate updates are hidden; finds that hiding states does not always improve privacy.

ID-6 *A Benchmark for Semantic Sensitive Information in LLMs’ Outputs*

Builds SemSI-Set and SemSI-Bench to measure semantic-level sensitive-information leakage in 25 SOTA LLMs; reveals that even simple natural questions can induce unintended disclosures.

ID-7 *Why In-Context Learning Models are Good Few-Shot Learners?*

Analyses transformer in-context learning as an implicit data-dependent algorithm beyond classical meta-learners and proposes meta-level curriculum strategies that further boost few-shot generalization.

ID-8 *Bridging Compressed Image Latents and Multimodal Large Language Models*

Presents a lightweight transform-neck and surrogate loss to adapt compressed image latents for downstream MLLM vision tasks without fine-tuning the main model, delivering strong rate-accuracy trade-offs.

C Preliminary Downstream Experiment: Four-Way Accept/Reject Classification

This experiment assesses whether upstream Bloom-taxonomy Q&A training benefits a downstream peer-review decision task. We map each paper’s average reviewer score from OpenReview to four discrete classes: **Reject** (1.0–2.5), **Weak Reject** (2.5–5.0), **Weak Accept** (5.0–7.5) and **Accept** (7.5–10.0). Then we pose it as a multiple-choice question with those four options.

Data and Models. We use 500 AI conference papers (ICLR&EMNLP 2024) with normalized 1–10 ratings, split 1985/307 train/test. All models employ the same **Qwen2.5-7B** backbone used in the rest of the paper:

- **Baseline (Instruction-only):** no Bloom Q&A pre-training, fine-tuned only on the 4-way labels.
- **Q&A + SFT $\rightarrow$ Instr.:** first supervised fine-tuned on Bloom QA, then instruction-tuned on 4-way labels.
- **Q&A + GRPO $\rightarrow$ Instr.:** first GRPO-trained on Bloom QA, then instruction-tuned on 4-way labels.

To mitigate catastrophic forgetting of the GRPO-aligned policy, we performed instruction tuning via LoRA adapters rather than full-parameter updates.

Results. Table 3 reports test-set accuracy with 95% bootstrap confidence intervals. All intervals overlap heavily, showing no discernible impact from upstream Q&A training.

Table 3: Four-way accept/reject classification on the held-out test set ($n = 307$). 95% bootstrap CIs shown.

| Model | Accuracy (%) |
|---------------------------------|----------------|
| Baseline (Instruction-only) | 44.0 ± 1.3 |
| Q&A + SFT $\rightarrow$ Instr. | 44.2 ± 1.2 |
| Q&A + GRPO $\rightarrow$ Instr. | 44.9 ± 0.9 |

Discussion. The nearly identical accuracies and overlapping confidence intervals indicate that upstream Bloom-taxonomy Q&A pre-training (whether SFT or GRPO) has no practical effect on this four-way peer-review classification task under our current dataset size and protocol, though we see marginal improvements in GRPO compared to SFT and baseline. Future work might explore larger corpora, alternative multi-task objectives, or chain-of-thought supervision to better transfer the benefits of structured QA training to decision-making scenarios.

Limitations. First, there was a scale mismatch between pretraining and downstream finetuning. Our Bloom-taxonomy QA training used only 3,543 questions drawn from 10 papers, which is a relatively small and domain-narrow corpus. In contrast, the downstream classification task fine-tunes on over 2,000 diverse conference papers. Any subtle reasoning benefits conferred by Bloom QA may be overwhelmed by this much larger instruction-tuning phase, masking their effect in

downstream evaluation. Future work should explore Bloom-style QA at scale and assess whether broader pretraining alters the final model’s decision patterns. Furthermore, the task at hand may have limited complexity. Mapping a numeric rating to one of four buckets may not demand the sort of multi-step or higher-order reasoning that Bloom-level QA might strengthen. Finally, there may be a potential model format mismatch. The model was originally trained to select the best multiple-choice answer from short science questions, whereas the peer-review classification is a more holistic textual assessment problem. This mismatch can reduce the direct benefit of prior QA training.